Title : AI-Based Property Buy Worthiness Assessment and Property Value Forecasting Using Location, Market, Builder and Civic Data (Property Buy / Not Buy Recommender)

Aim : To develop a practical, privacy-preserving AI/ML decision-support system that evaluates whether a user-specified property in Jammu is worth buying and forecasts its potential future value. The system integrates public real-estate datasets, limited web-scraped civic and builder indicators, and a Retrieval-Augmented Generation (RAG) agent for transparent builder-credibility synthesis. It prioritises data-driven transparency, explicit bias mitigation, and client-side execution so that no user data leaves the device.

Introduction:

Property purchase is one of the major financial decisions made by individuals and families. Unlike ordinary consumer purchases, evaluating a property requires simultaneous consideration of current price, historical price trends, location characteristics, infrastructure development, builder credibility, civic conditions and future growth potential.

Existing real-estate platforms primarily facilitate property discovery by providing listings, prices, locations and basic characteristics. However, the information required to determine whether a specific property is actually worth buying is often distributed across different sources and difficult to evaluate collectively. Historical property prices may indicate market appreciation, infrastructure development may influence future demand, civic complaints can signal locality-level issues, and builder credibility can indicate project-related risk.

The proposed project develops an AI/ML-based property decision-support system that evaluates a given property using multiple relevant indicators rather than simply recommending properties based on user preferences. The system focuses on property assessment rather than property recommendation: it evaluates a property provided by the user. All inference is performed client-side, ensuring that no user data leaves the device.

Abstract

Property purchase is a high-stakes financial decision requiring simultaneous evaluation of price, market history, locality conditions, legal status, civic risk and builder credibility, yet in Jammu this information is scattered across unconnected sources. This project proposes an **AI-Based Property Buy-Worthiness Assessment and Property Value Forecasting System** that evaluates a user-specified property rather than recommending listings.

The system uses only public data: (i) publicly displayed Jammu listings on 99acres (1,200 records, 36 localities) for prices and property attributes; (ii) **Jammu Development Authority (JDA)** layout approvals and master-plan records for approval status and land use; (iii) **J&K RERA** registers for project registration, builder history and complaints; (iv) **Jammu Municipal Corporation** and Smart City Jammu records for civic complaints, water, sewage and road data; (v) J&K Revenue and Registration Department records and circle rates for ownership and registry type; (vi) J&K SDMA flood-hazard maps; and (vii) OpenStreetMap for distances, schools, hospitals and transport. A local Retrieval-Augmented Generation

(RAG) component synthesises builder credibility from public RERA and news documents.

These sources yield 35 variables (28 base, 7 engineered). An ensemble of Random Forest, XGBoost and LightGBM forecasts 1-, 3- and 5-year values. Buy-worthiness is a transparent weighted composite: **Score = 100 × Σ $w_i \cdot I_i$**, where each $I_i \in [0,1]$ is a min–max-normalised index built from related variables: valuation fairness (25%; price-versus-locality and price-versus-type ratios, locality price rank), growth outlook (25%; past growth, forecast growth, growth momentum), legal clarity (20%), builder credibility (10%), location and amenity access (10%), and inverted civic risk (10%). Scores ≥70 indicate Buy, 40–69 Maybe and <40 No-Buy; missing JDA approval or an unclear registry caps the verdict at Maybe. Initial weights are expert-defined, then validated against SHAP importances and stratified cross-validated performance, while a classifier supplies the final label with confidence.

Explainable AI shows each factor's contribution to every verdict, so buyers can see why a property is rated as it is. Evaluation uses R^2, MAE, RMSE, MAPE, accuracy and F1, with bias audits across localities and price bands. All inference runs client-side, so no user data leaves the device, in line with the V.E.T.S. framework.

Problem Statement:

A property buyer typically has access to basic information such as property price, size, location and builder name, but determining whether the property represents a worthwhile purchase requires analysis of several additional heterogeneous factors.

Historical price trends can indicate the behaviour of a local property market, while infrastructure development can influence future appreciation. Builder credibility can provide an indication of project-related reliability, whereas civic complaints can highlight potential issues within a locality. These factors are often distributed across different sources and difficult to evaluate collectively.

Therefore, there is a need for an AI/ML-based decision-support system that can evaluate the buy-worthiness of a specified property and forecast its potential future value by integrating property, market, locality, builder, civic and infrastructure-related indicators derived from public datasets. The system must remain feasible within a minor-project timeline, incorporate modern retrieval-augmented reasoning for builder analysis, and explicitly address data privacy and historical bias risks.

Literature review:

Existing research demonstrates the effectiveness of machine learning techniques for property valuation and real-estate price prediction. Potrawa and Tetereva proposed a machine-learning-driven hedonic pricing model for the real-estate market, demonstrating how property characteristics and contextual factors can be incorporated to estimate property prices [1].

Rico-Juan and Taltavull de La Paz investigated machine learning and spatial hedonic approaches for analysing housing prices, emphasising the importance of location-related characteristics and model explainability [2].

Hernes et al. investigated machine-learning approaches for real-estate valuation using historical transaction data and compared different predictive models [3].

Mathotaarachchi et al. presented a review of advanced machine-learning techniques for predictive modelling of property prices, highlighting the importance of selecting suitable techniques and evaluating performance [4].

Foryś examined machine-learning approaches for house-price analysis by comparing regression-based models with neural-network approaches [5].

Baur, Rosenfelder and Lutz investigated automated real-estate valuation using machine-learning models and property descriptions [6].

Recent research in the Indian context has explored Linear Regression, Support Vector Regression, K-Nearest Neighbours, Random Forest and XGBoost for predicting residential property prices in Mumbai [7].

These studies establish the usefulness of machine learning for property valuation, historical price analysis and location-based prediction. The proposed system extends this foundation by incorporating builder credibility (via a RAG agent), civic indicators, infrastructure development and locality scores, while combining property assessment with future-value forecasting and ensuring full client-side privacy and bias transparency.

Tech Stack :

Category	**Proposed Technology**
Programming	Python
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn
Advanced ML	XGBoost, LightGBM, Random Forest

RAG Component	Sentence-Transformers + FAISS/Chroma (local)
Visualization	Matplotlib, Plotly, Chart.js, Leaflet.js
Database	SQLite (local) / embedded CSV
Interface	Streamlit + Vanilla JavaScript (client-side)
Development	Jupyter / Google Colab, VS Code
Version Control	Git / GitHub

Objectives :

1. To collect and integrate property, land-transaction, locality, infrastructure and civic-complaint data exclusively from publicly available sources.
2. To perform a documented data-preprocessing and feature-engineering pipeline (missing-value treatment, encoding, robust scaling, outlier handling, leakage checks) to derive meaningful indicators related to property price, locality quality, market growth and development.
3. To develop a property buy-worthiness model that evaluates whether a given property is financially and contextually attractive based on multiple factors and outputs a Buy / No-Buy / Maybe score with confidence.
4. To develop a property-value forecasting model (ensemble of Random Forest, XGBoost and LightGBM) capable of estimating future property values using historical price trends and development-related indicators.
5. To incorporate builder/developer credibility as a measurable feature using a lightweight RAG agent that synthesises publicly available project-history and complaint summaries.
6. To analyse the influence of infrastructure development and locality characteristics on property-value appreciation through feature-importance and explainability techniques.
7. To provide interpretable results showing the major factors contributing to the property's assessment and predicted value, together with explicit bias audits across price bands and localities.
8. To evaluate the developed models using appropriate machine-learning and forecasting

metrics (R^2, MAE, RMSE, MAPE, accuracy, F1) and to ensure 100 % client-side inference for privacy.

Dataset:

The dataset consists of 35 variables consisting of 28 base variables and 7 engineered variables .The variables are also categorized as below:

A. Property:

1. Property_type
2. Area_sqft
3. Price_per_sqft
4. Property_age_years
5. Corner_plot
6. Facing_direction
7. Road_width_ft

B. Location & Access:

8. locality (36 levels)
9. Distance_to_city_center_km
10. Distance_to_highway_km
11. Locality_score
12. Infrastructure_score
13. Road_connectivity_score
14. Public_transport_score

C. Amenities:

15. Nearby_schools_count
16. Nearby_hospitals_count

D. Legal & builder:

17. Ownership_type
18. Registry_type
19. Jda_approved
20. Bank_loan_approved
21. Builder_name
22. Builder_credibility_score

E. Civic & utilities:

23. Civic_complaints_count
24. Flood_risk
25. Electricity_connection
26. Sewage_connection
27. Water_supply_score

F. Market history:

28. Past_5yr_price_growth_pct

G. Engineered:

29. Price_vs_locality_type (asking price ÷ median of same type in same locality)
30. Price_vs_type (asking price ÷ citywide median of same type)
31. `Growth_momentum (forecast growth − past growth)`
32. Legal_clarity_index (registry, ownership, JDA, loan eligibility)
33. Amenity_access_index (schools, hospitals, transport, distance to centre)
34. Civic_risk_index (complaints, flood, water, sewage, electricity)
35. locality_price_rank (locality median price percentile)

Sources: 99acres , Realtor.com



Methodology:

1. **Data Collection** – Gather only public datasets: verified Jammu property listings (1,200 records across 36 localities), open civic and infrastructure indicators, and limited publicly available builder project histories.
2. **Data Preprocessing** – Clean, integrate and transform data; handle missing values, encode categoricals, apply robust scaling and outlier capping; document every step.
3. **Exploratory Data Analysis** – Visualise distributions, identify key factors and analyse correlations to guide feature engineering.

4. **Feature Engineering** – Create location-based, price-trend, builder-credibility, civic-complaint, infrastructure-development and locality-score features (35 engineered variables). 5. **Model Development** –

- Price forecasting: weighted ensemble (Random Forest + XGBoost + LightGBM).
- Buy-worthiness: multi-class / thresholded classification head.
- RAG agent: local vector store + retrieval + lightweight synthesis for builder-credibility reports.

6. **Model Evaluation** – Regression metrics (R^2, MAE, RMSE, MAPE) and classification metrics (accuracy, F1, confusion matrix) with stratified cross-validation.
7. **Investment Scoring** – Calculate weighted buy-worthiness score and classify into Buy / No-Buy / Maybe.
8. **Price Forecasting** – Generate 1-year, 3-year and 5-year value estimates together with expected appreciation percentage.
9. **Explainability & Bias Mitigation** – Feature-importance visualisations, SHAP-style explanations and performance audits across price ranges and localities.
10. **User Interface & Output** – Streamlit / client-side JavaScript application presenting property details, buy-worthiness score, forecast values, builder-credibility report, feature contributions and bias notes. Continuous improvement via periodic public-data updates and model retraining.

V.E.T.S. Framework for Your Project

V — Viability

What makes the project feasible?

- Uses publicly available property, civic, infrastructure and builder-related data.
- Dataset currently contains **1,200 Jammu property records across 36 localities**.
- Uses an achievable technology stack: **Python, Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, Random Forest, Sentence-Transformers, FAISS/Chroma and Streamlit**.
- Uses local SQLite/CSV storage and client-side inference, reducing infrastructure requirements.
- Modular development can follow the proposed **16-week, four-phase roadmap**.

VETS statement:

> **V — Viability:** The system is feasible through publicly available Jammu property, civic, infrastructure and builder data, a manageable 35-variable feature set, locally deployable ML/RAG components, and a modular 16-week implementation roadmap.

E — Engineering Depth

This is where your project becomes more than a normal property website.

Your system contains:

Data Engineering

→ Data collection
→ Cleaning
→ Missing-value treatment
→ Encoding
→ Outlier handling
→ Feature engineering

35-Variable Feature Layer

→ Property
→ Location & access
→ Amenities
→ Legal & builder
→ Civic & utilities
→ Market history
→ Engineered indicators

AI/ML Layer

- Random Forest
- XGBoost
- LightGBM
- Buy / No-Buy / Maybe classification
- 1-, 3- and 5-year value forecasting

RAG Layer

- Sentence-Transformer embeddings
- Local FAISS/Chroma vector store
- Retrieval of builder/project information
- Builder credibility synthesis

Explainability Layer

- Feature importance
- SHAP-style explanations
- Bias audits across localities and price bands

So your VETS engineering statement can be:

> **E — Engineering Depth:** The system integrates heterogeneous data engineering, 35-variable feature engineering, ensemble machine learning, RAG-based builder credibility analysis, future-value forecasting, explainable AI and bias auditing rather than functioning as

a simple property-listing interface.

This is strongly supported by your existing methodology.

T — Trend Alignment

For **your particular project**, I would use:

Edge Computing + Privacy-Preserving AI

rather than trying to add blockchain, Web3 or unnecessary multi-agent architecture.

Your project already specifies **client-side inference so that user data does not leave the device**.

This gives you a genuine trend-alignment argument:

> **T — Trend Alignment:** The system adopts privacy-oriented edge/client-side AI by performing inference locally, while incorporating a local Retrieval-Augmented Generation pipeline for dynamic builder-credibility synthesis and explainable AI for transparent decision support.

You can also mention **RAG** as the modern AI component:

RAG → Builder documents → Retrieval → Relevant evidence → Credibility synthesis → Assessment

Your VETS document lists Edge Computing as one of the possible trend-alignment directions, so this fits the framework without artificially changing your architecture.

S — Social / Industrial Impact

Your project addresses a significant problem in property purchasing: information is distributed across property listings, price histories, civic information, infrastructure indicators and builder information. Your original document explicitly identifies this fragmentation as a problem.

Impact areas

For property buyers

- Understand whether a specified property is worth buying.
- Identify price discrepancies.

- View future-value estimates.
- Understand locality and civic risks.
- Examine builder credibility.

For transparency

- Shows factors contributing to the assessment.
- Provides explainable predictions.
- Provides confidence information.
- Performs bias audits.

For privacy

- User property information can be processed locally.
- No dependence on sending personal inputs to a central server.

For the real-estate domain

- Combines property, market, civic, infrastructure and builder information into one analytical system.

Future Enhancement (Quality of Life Index):

A multi-dimensional Quality of Life Index for each locality will be developed using publicly available or openly licensed indicators: Air Quality Index (AQI), ambient noise levels, peak-hour traffic congestion, crime-report index, water-supply and quality data, electricity reliability, proximity to the nearest hospital, and monthly property-registry activity. This index will further enrich locality scoring and investment recommendations.

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